

Verification Continuum™

Verdi Installation and System Administration Guide

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1

About This Document

In this chapter:

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Purpose

This document explains how to install the Verdi and Siloti systems, symbol libraries and memory definition table (MDT) files. The manual should be read from beginning to end, although you may skip any sections with which you are already familiar.

For detailed descriptions of individual commands, refer to the appropriate chapter of the *Verdi and Siloti Command Reference Manual*. For detailed usage on any of the installed products, refer to the appropriate user's guide as listed in the *Related Publications* section of this chapter.

Audience

The audience for this document includes engineers who are familiar with languages and tools used in design and verification such as Verilog, VHDL, and SystemVerilog, and typically install and set-up tool software.

This document assumes that you have a basic knowledge of the platform on which your version of the Verdi and Siloti system runs: UNIX or Linux and that you are knowledgeable in Verilog or VHDL and digital logic design.

Book Organization

This *Installation and System Administration Guide* is organized as follows:

- *About This Document* provides an introduction to this document and explains how to use it.
- *Installing Verdi and Siloti Software* explains how to install Verdi and Siloti systems on the platforms they support.

Conventions Used in This Book

The following conventions are used in this document:

- *Italics font* is used for emphasis, book titles, section names, design names, file path, and file names within paragraphs.
- **Bold** is used to emphasize text, highlight titles, menu items, and other Verdi terms.
- `Courier type` is used for program listings. It is also used for test messages that the Verdi platform display on the screen.
- **NOTE** describes important information, warnings, or unique commands.
- **Menu -> Command** identifies the path used to select a menu command.
- Left-click or Click means click the left mouse button on the indicated item.
- Middle-click means click the middle mouse button on the indicated item.
- Right-click means click the right mouse button on the indicated item.
- Double-click means click twice consecutively with the left mouse button.
- Shift-left-click means press and hold the <Shift> key then click the left mouse button on the indicated item.
- Drag-left means press and hold the left mouse button, then move the pointer to the destination and release the button.
- Drag means press and hold the middle mouse button on the indicated item then move and drop the item to the other window.

Related Publications

- *Verdi and Siloti Command Reference Manual* - provides detailed information on the Verdi and Siloti command sets.
- *Verdi and Siloti Quick Reference Guide* - provides a quick reference for using the Verdi and Siloti systems with typical debug scenarios.
- *Linking Novas Files with Simulators and Enabling FSDB Dumping* - provides detailed information on linking Novas object files with supported simulators for FSDB dumping and the related dumping commands.
- *Verdi User Guide and Tutorial* - provides detailed information on using the Verdi platform.
- *Siloti User Guide and Tutorial* - provides detailed information on using the Siloti system.
- *nAnalyzer User Guide and Tutorial* - provides detailed information on using the *nAnalyzer* Design Analysis module.
- *nECO User Guide and Tutorial* - provides detailed information on using the *nECO* Automated Netlist Modification module.
- *Release Notes* - provides current information about the latest software version. Refer to the 'View release notes' link on the product downloads page.
- Language Documentation

Hardware description (Verilog, VHDL, SystemVerilog, etc.) and verification language reference materials are not included in this manual. For language related documents, refer to the appropriate language standards board (www.ieee.org, www.accellera.org) or vendor (www.synopsys.com, www.cadence.com) websites.

2

Installing Verdi and Siloti Software

In this chapter:

- [Overview](#)
- [Installing Verdi and Siloti on UNIX Platforms](#)
- [Setting Up the License Server](#)
- [Setting Up the Environment and Running the Software](#)

Overview

Workstation Requirements

Before installing Verdi and Siloti systems, verify that you have the following components:

- Software DVD or the software files from the Internet.
- 5.5 GB hard drive space (including downloaded files).
- The license file.

Supported Simulators

Refer to the “Novas Object File Cross Reference Matrix” section in the *Linking Novas Files with Simulators and Enabling FSDB Dumping* document for the current list of supported simulator versions.

Linking Novas Object Files

For complete details on linking the Novas object files for FSDB dumping with various simulators, see the *Linking Novas Files with Simulators and Enabling FSDB Dumping* document.

Installing Verdi and Siloti on UNIX Platforms

Supported Platforms

Verdi and Siloti systems support most major Sun and Linux workstations. Refer to the *Release Notes* for the current list of platforms and OS versions.

For detailed platform support information, see the **Release Specific Support** page on **Synopsys Qualified System Configuration** web site at the following address:

<http://www.synopsys.com/qsc>

This web page provides release-specific information about supported hardware, operating systems, and required operating system patches. If the required patch described on this page is not available from the platform vendor, install the most recent patch instead.

Note:

Solaris platform is no longer supported.

Installing from the Internet

Perform the following steps to install from the Internet:

1. On the Synopsys SolvnetPlus website, enter your User ID and Password, and then click **SIGN IN**.
2. On the top bar, click the **Downloads** link.
3. Under **My Product Releases**, scroll and click **Verdi3**.
4. Click the version of interest and then click the **Download Here** link.
5. Click the **YES, I AGREE TO THE ABOVE TERMS** button.
6. Select the required files by clicking the appropriate **Download** button.
 - Common Files
 - Executable Files for your specific architecture(s) e.g. 32/64 Bit Linux
7. Invoke the Synopsys Installer by using one of the following commands:
 - > %installer #To use a text script
 - > %installer -gui #To use a GUI

You must install the latest release in an empty directory using the latest version of the Synopsys Installer. Do not install this release over an existing release of a Synopsys tool.

For information about downloading the Synopsys Installer, see the Synopsys Installer section at the following address:

<http://www.synopsys.com/install>

The following directories are created:

bin	The executable programs
license	An empty directory
etc	Software default-setting files
doc	Documentation including online help, release notes, and so on
share	Symbol libraries, NPI libraries, ovm/uvm libraries, PLI modules, PLI directory
demo	Demo samples
platform	Platform-dependent data

Setting Up the License Server

Before running Synopsys tools, the Synopsys Common Licensing (SCL) software must be installed and configured, the license key file must be retrieved, and the license file environment variable must be defined. For detailed information about SCL installation and setup, see the *Synopsys Licensing Quickstart Guide*.

Note:

This release of the product uses the SCL daemon named *snpslmd*. To use the product, SCL 11.6 or higher versions of the *snpslmd* daemon must be used with the latest license key file. Refer to the *SCL 11.6 Release Notes* and

Administration Guide for further details. A copy of the documentation can be found in the following address: <http://www.synopsys.com/licensing>

The following table summarizes the supported versions of Verdi and the license files:

Verdi Version	License file
Before Verdi 2014.03	CVD phase-3 license file
Verdi 2014.03 and later	CVD phase-3 license file (or) phase-4 license file (pure Synopsys keys)

Setting Up the Environment and Running the Software

1. If needed, specify the symbol libraries.
2. Confirm Perl (or a link to the executable) is located in `/usr/local/bin` and that it exists in your path.

Note:

If Perl is not in this location, some utilities may not work correctly.

3. Set the VERDI_HOME environment variable to the installation directory.

```
% set VERDI_HOME <VERDI_INST_DIR>
```

4. Add the Verdi application (binary) to the search path:

For the C (csh) shell:

```
% set path = ($VERDI_HOME/bin $path)
```

For Bourne (sh) and Korn (ksh) shells:

```
% PATH=$VERDI_HOME/bin:$PATH
```

5. Run the software.

Use the following command to run the Verdi and Siloti system:

```
> % verdi
```

Use the following command to run *nWave* standalone:

```
> % nWave
```

Refer to the *Verdi and Siloti Command Reference Manual* for complete details on executing the Verdi and Siloti platform or *nWave* (verdi and nWave utility descriptions in the *Utilities* chapter).

6. Optionally, run the *nCompare* software.

Use the following command to run the *nCompare* module in the batch mode:

```
> % nCompare [batch mode options]
```

Use one of the following commands to run *nCompare* in GUI mode:

```
> % nCompare -gui [rule file | error report file] or  
> % nCompareg [rule file | error report file]
```

Refer to the *nCompare* chapter in the *Verdi and Siloti Command Reference Manual* for details.

3

Installing Verdi Supplementary Package

In this chapter:

- [Overview](#)
- [Installing in GUI Mode](#)
- [Installing in Batch Mode](#)

Overview

Workstation requirements before installing Verdi Supplementary package, verify that you have the following components:

- Software files from the Synopsys SolvnetPlus website.
- 35 GB hard drive space (including downloaded files).
- The license file.

If VCS is installed

You can use Vericom/Vhdlcom Advanced Mode without by simply setting up `VCS_HOME` and point to the VCS installed directory.

```
% setenv VCS_HOME <VCS_installed_directory>
```

If VCS is not installed

You need to install Verdi Supplementary package under path such as:

```
$VERDI_HOME/platform/
```

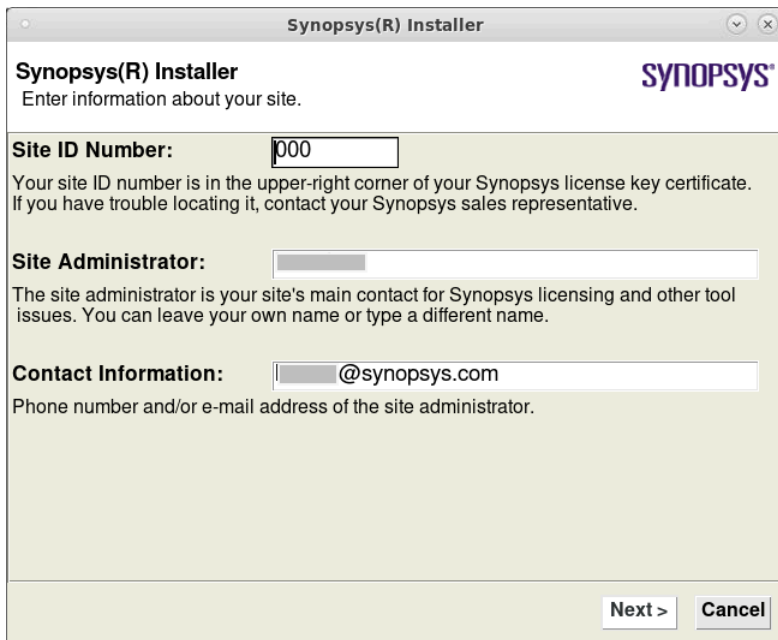
Installing in GUI Mode

Install Verdi first, if you have not installed it

Note:

The images in this chapter are for reference purpose. The product version in the images may differ based on the actual release version.

1. Enter the basic information.



The image shows the 'Synopsys(R) Installer' window. The title bar says 'Synopsys(R) Installer'. The main window has a header with 'Synopsys(R) Installer' and the Synopsys logo. Below the header, it says 'Enter information about your site.' There are three input fields: 'Site ID Number:' with the value '000', 'Site Administrator:' with a greyed-out field, and 'Contact Information:' with the value '@synopsys.com'. Below each field is a descriptive text. At the bottom right, there are 'Next >' and 'Cancel' buttons.

Synopsys(R) Installer
Enter information about your site.

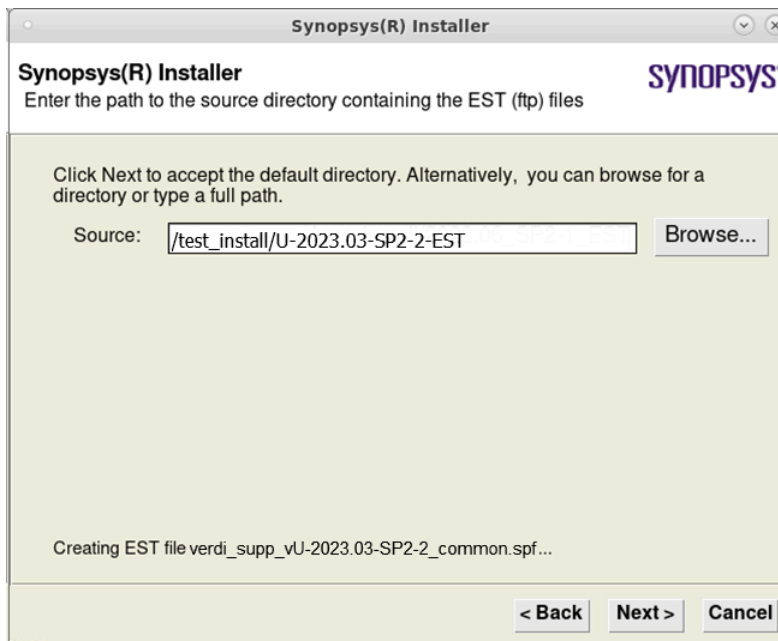
Site ID Number: 000
Your site ID number is in the upper-right corner of your Synopsys license key certificate. If you have trouble locating it, contact your Synopsys sales representative.

Site Administrator:
The site administrator is your site's main contact for Synopsys licensing and other tool issues. You can leave your own name or type a different name.

Contact Information: @synopsys.com
Phone number and/or e-mail address of the site administrator.

Next > Cancel

2. Select the source directory.



The image shows the 'Synopsys(R) Installer' window. The title bar says 'Synopsys(R) Installer'. The main window has a header with 'Synopsys(R) Installer' and the Synopsys logo. Below the header, it says 'Enter the path to the source directory containing the EST (ftp) files'. There is a text area with instructions: 'Click Next to accept the default directory. Alternatively, you can browse for a directory or type a full path.' Below this, there is a 'Source:' label, a text input field with the value '/test_install/U-2023.03-SP2-2-EST', and a 'Browse...' button. At the bottom, there is a status bar that says 'Creating EST file verdi_supp_vU-2023.03-SP2-2_common.spf...'. At the bottom right, there are '< Back', 'Next >', and 'Cancel' buttons.

Synopsys(R) Installer
Enter the path to the source directory containing the EST (ftp) files

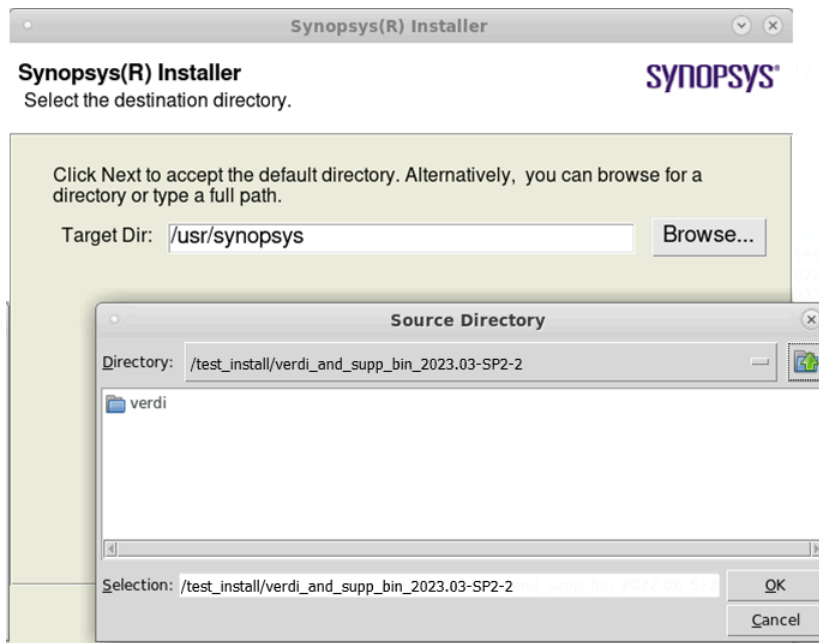
Click Next to accept the default directory. Alternatively, you can browse for a directory or type a full path.

Source: /test_install/U-2023.03-SP2-2-EST Browse...

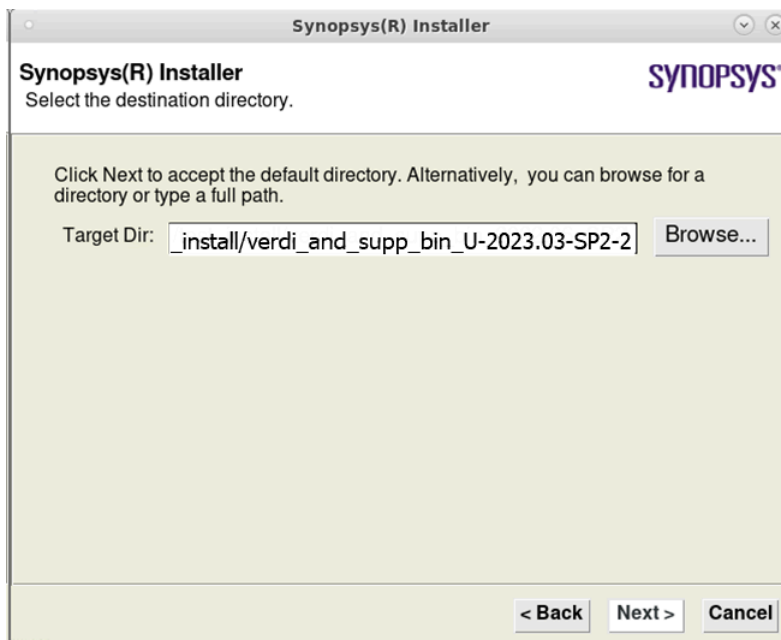
Creating EST file verdi_supp_vU-2023.03-SP2-2_common.spf...

< Back Next > Cancel

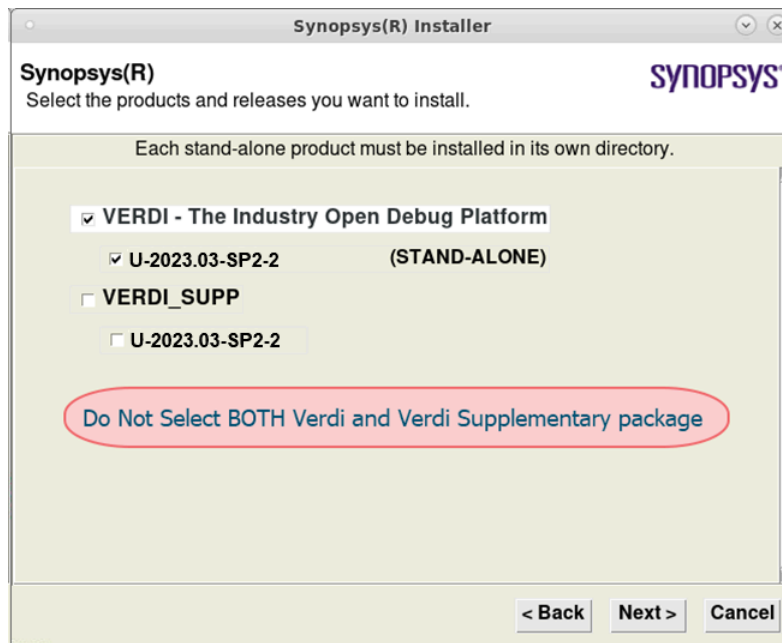
3. Select the target directory.



4. Click **Next**.



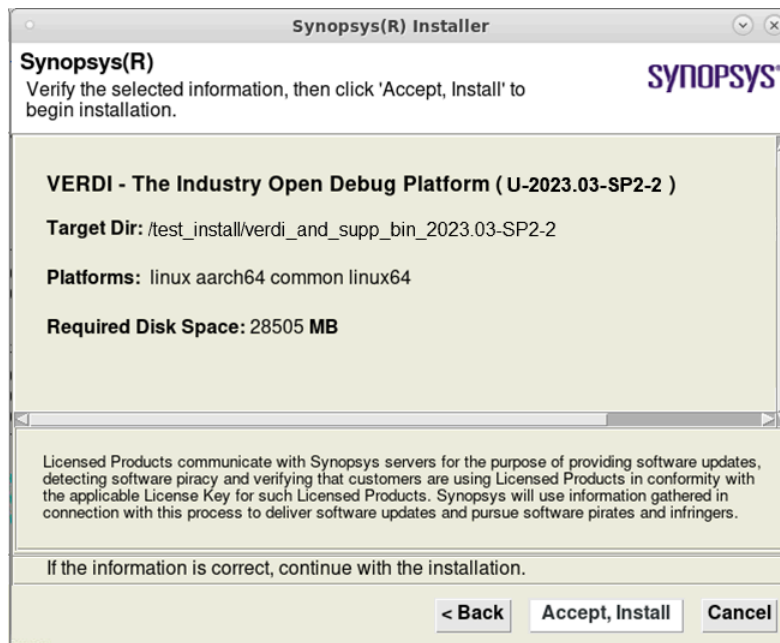
5. Select Product **VERDI-The Industry Open Debug Platform**.



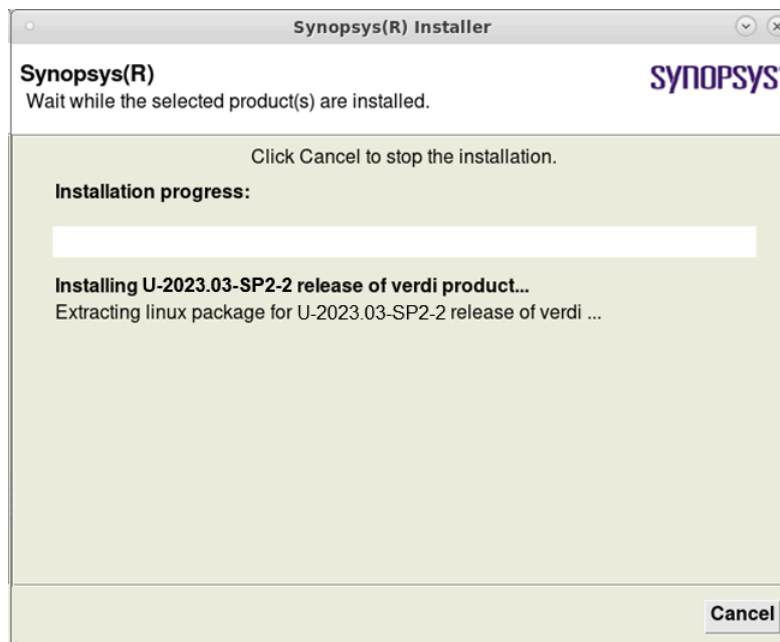
6. Select all platforms.



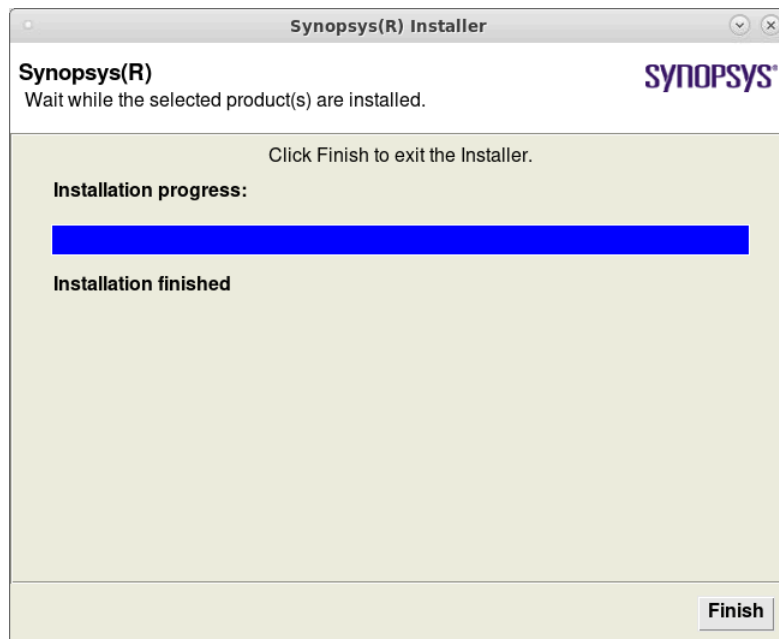
7. Check the installation configurations, and then click **Accept, Install**.



Installation is in progress.

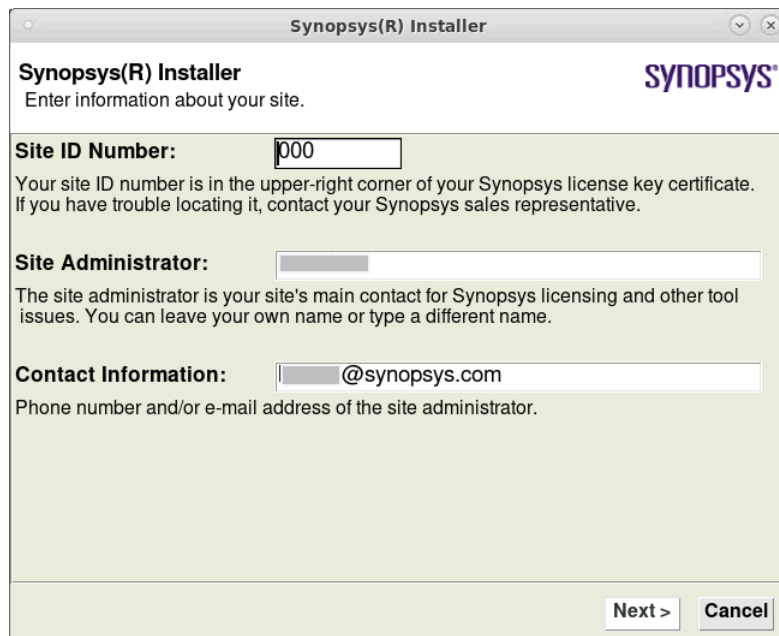


Installation is completed.

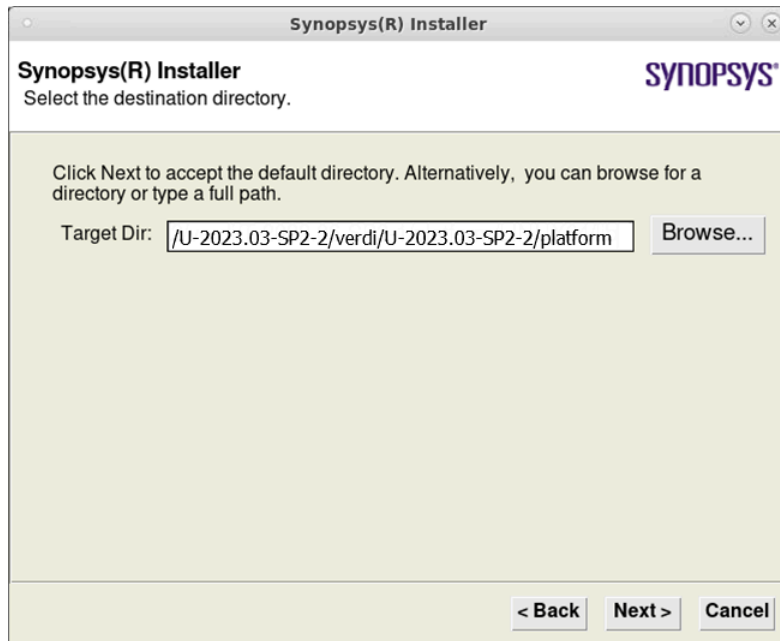


Install the Verdi Supplementary Package

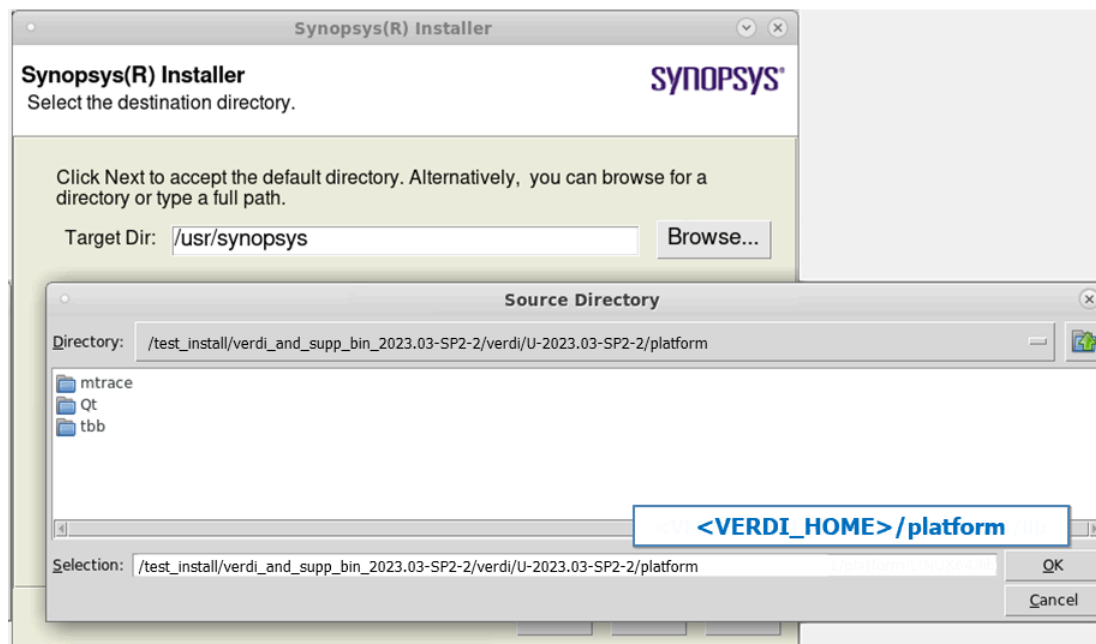
1. Enter the basic information.



2. Select the source directory.

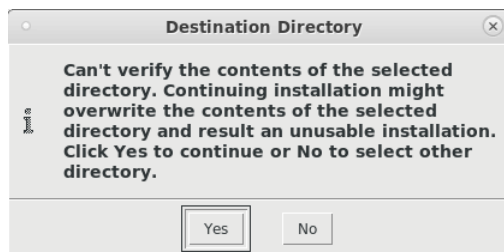


3. Select the target directory `<VERDI_HOME>/platform/`.

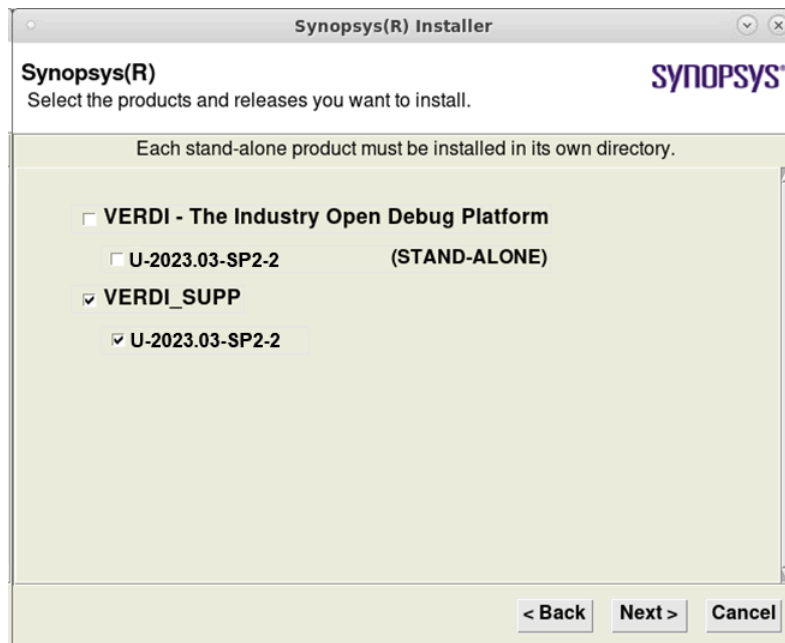




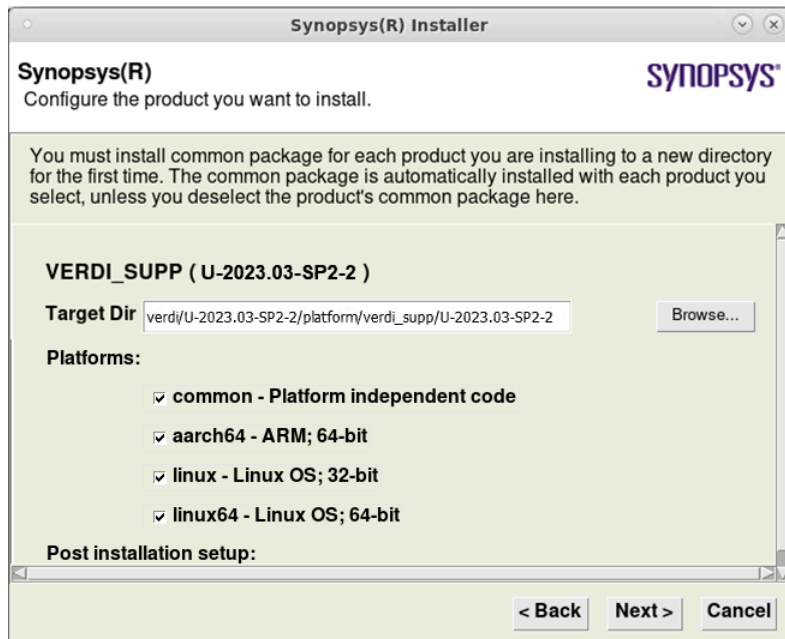
4. Ignore the following warning message:

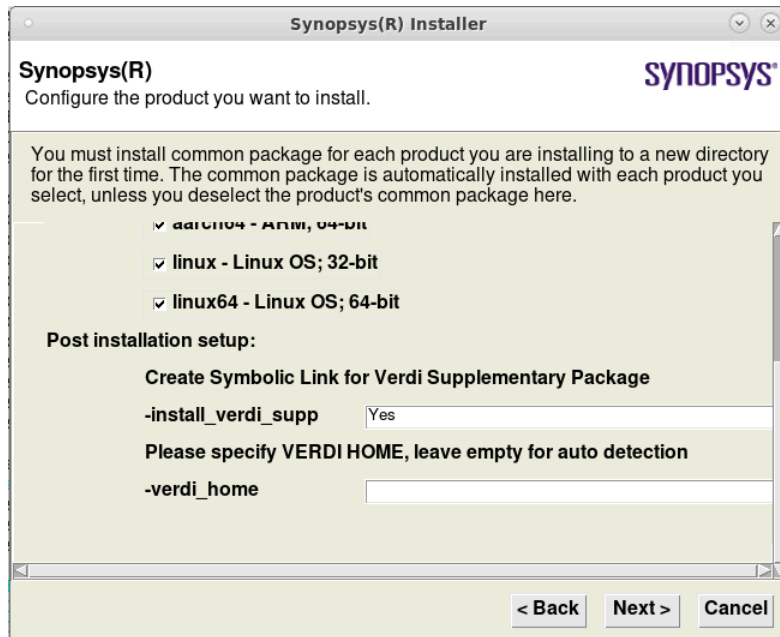


5. Select Product **VERDI_SUPP**.

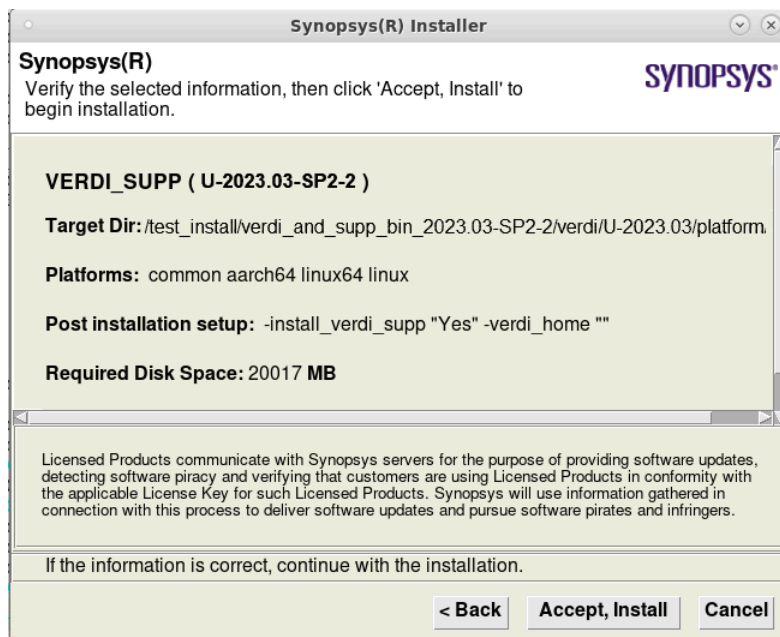


6. Retain the default settings.

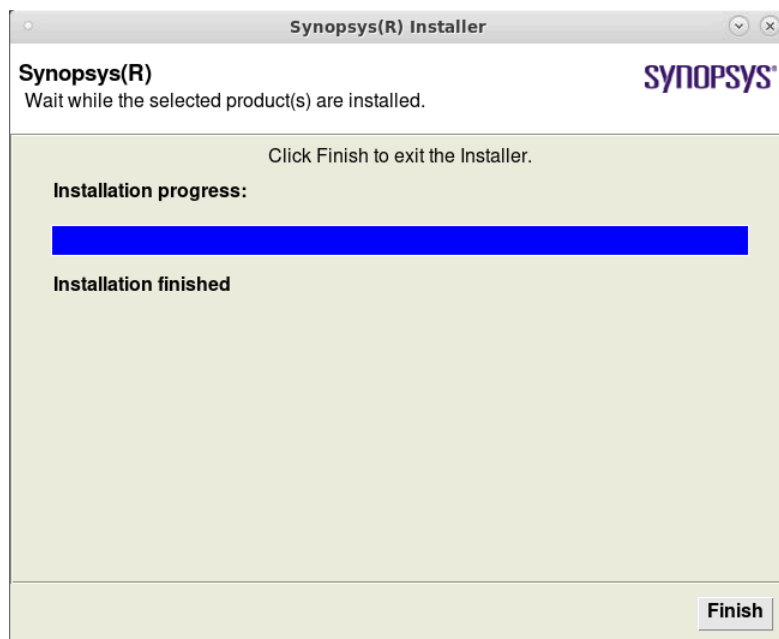




7. Check the installation configurations, and then click **Accept, Install**.



Installation is completed.



After installation

You will see the folder **verdi_supp** and a symbolic link as follows:

```
> ls <VERDI_HOME>/platform/LINUX64/lib/  
mtrace Qt tbb vcs verdi_supp
```

Installing in Batch Mode

Enter information for installer questions:

```
Enter the path to the source directory containing the downloaded EFT
file(s) [<CWD>]: <Where you store the spf files of Verdi Supplementary
Package>
Enter the full path to the directory where you want to install Synopsys
products. If the directory does not exist, it will be created.
[/usr/synopsys]: <VERDI_HOME>/platform/LINUX64/lib/
Creating EST file verdi_supp_<version>_common.spf....
Creating EST file verdi_supp_<version>_common.spf successfully
Extracting release information from verdi_supp_<version>_common.spf
Extracting release information from verdi_supp_<version>_common.spf
successfully
Your site ID number is in the upper-right corner of your Synopsys license
key certificate. If you have trouble locating it, contact your Synopsys
representative.
Site ID number [000]: <Your Site ID>
Post installation setup for VERDI_SUPP:
  Create Symbolic Link for Verdi Supplementary Package [Yes]: <Enter with
  default Yes>
....
If the information is correct, continue with the installation.
Accept, Install? [yes]: <Enter with default Yes>
INFO: Prepare for installing <version> release of product verdi_supp ...
INFO: Extracting common package from verdi_supp_<version>_common.spf ...
INFO: Verify SHA256 sum for verdi_supp.bz2 ....
INFO: Installing <version> release of product verdi_supp to <The full
path of target installation path> ....
INFO: Installing common package for <version> release of product
verdi_supp ...
[VERDI SUPPLEMENTARY INSTALL] VERDI_HOME is <VERDI_HOME>
[VERDI SUPPLEMENTARY INSTALL] cd <VERDI_SUPP_INSTALLATION_PATH>
[VERDI SUPPLEMENTARY INSTALL] Create symbolic link ...
[VERDI SUPPLEMENTARY INSTALL] ln -s <VERDI_SUPP_INSTALLATION_PATH>
verdi_supp/<version>
[VERDI SUPPLEMENTARY INSTALL] Completed!!!
```

After installation

You will see a folder **verdi_supp** and a symbolic link below:

```
> ls <VERDI_HOME>/platform/LINUX64/lib/
mtrace Qt tbb vcs verdi_supp
```